Silica commander AI (Sol / Centauri): refinery spam and missing HQ expansion

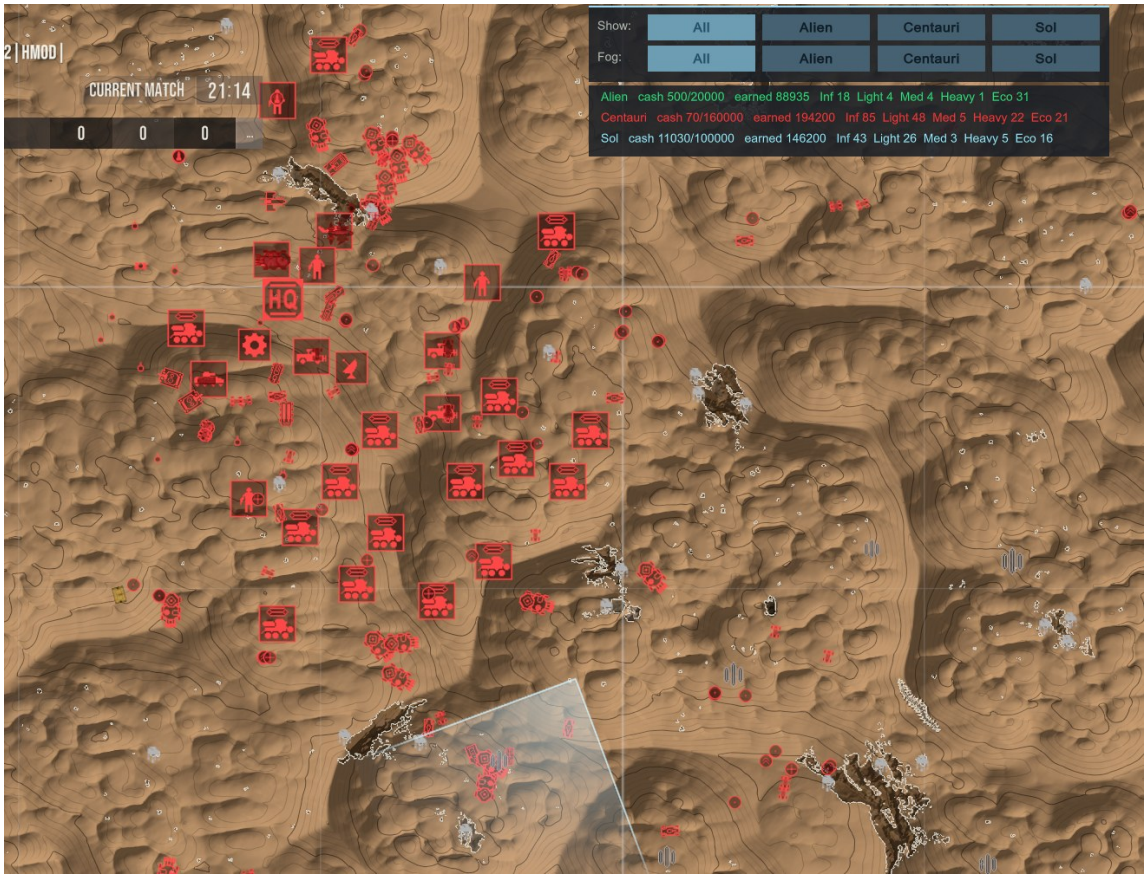
Field report from a modded public server, Release 1.0.15, 2026-11-01. Author: DrMuck. Code-level root causes and suggested fixes are in the companion review of the decompiled 1.0.9 AI (the economy code is unchanged through 1.0.15): https://claude.ai/artifact/NGgmrM5c7UxRWf4Loobwpz. This document shows what it looks like in play and summarises the behaviour we would like to see.

1. Summary

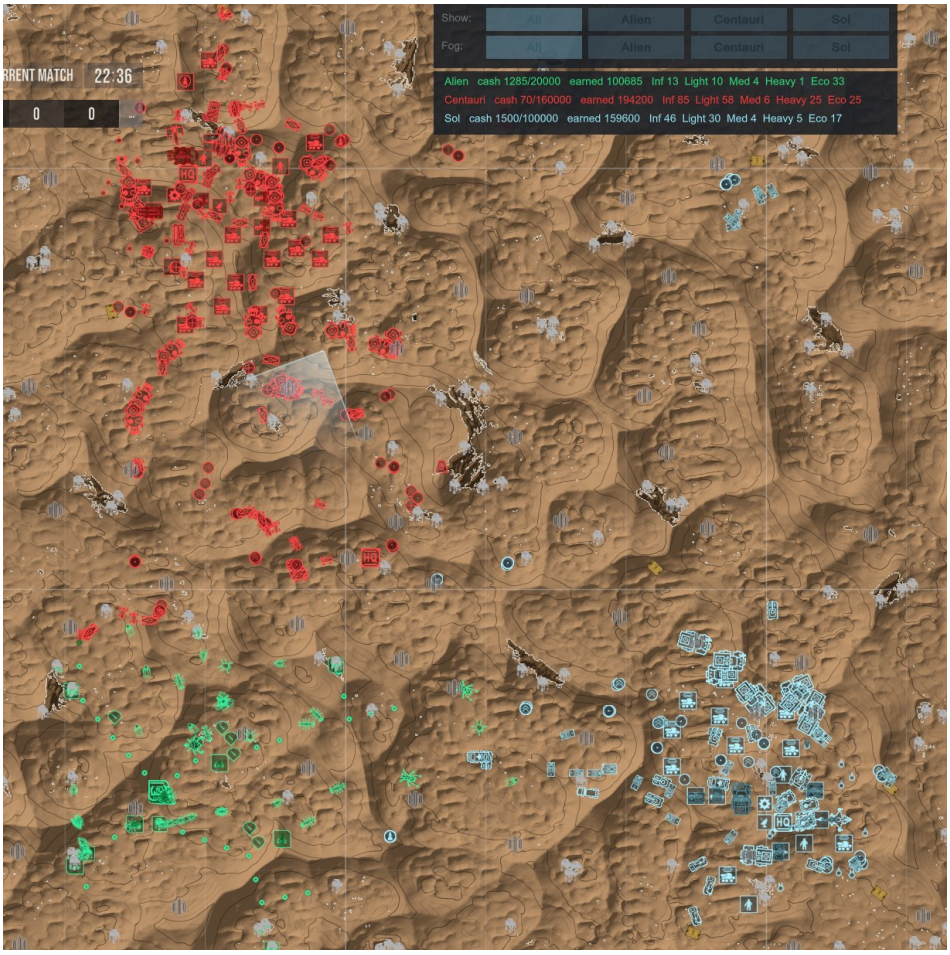
- **Problem 1, refinery spam.** The vanilla AI builds refineries without regard to balterium patches: several refineries per patch, stacked side by side, ramps not pointing at the field. Centauri is hit hardest because its refinery footprint is easier to place. In the 3-way round below Centauri had 21 refineries at 21 min and 25 at 22:36, Sol 16-17, with the Centauri cluster covering only a handful of patches.
- **Problem 2, almost no HQ expansion.** Once the first base's patches are taken, the AI should push a second HQ toward unclaimed fields. It very rarely does; the money goes into more refineries around the first HQ instead.
- **Asked for.** One refinery per balterium patch, built as close to the patch as the footprint allows with the ramp facing the patch; and an HQ-expansion reward that scores a candidate by the unclaimed patches it would bring into refinery reach, placed toward the far edge of the HQ-to-HQ reach. Both rules must derive every distance from the live ConstructionData, because build radii are modded on many servers (ours: refinery reach 1000 m instead of 600 m, HQ reach 2500 m instead of 1500 m, HQ at tier 4 instead of 5).
- **Reference.** Our own commander mod (Si_RTS_AI) implements both rules in a simple form; section 5 describes that logic and shows a Sol base built by it (screenshot 4) for comparison with the vanilla result.

2. Observed: refinery spam (vanilla AI, modded server)

Three-way match (Alien / Centauri / Sol); all three commanders are the vanilla AI. The server runs larger build radii (Refinery 1000 m, HQ 2500 m, HQ unlocks at tier 4), which the vanilla AI honours only partially (section 4).



Screenshot 1, 21:14. Centauri base (red). Overlay top right: Centauri cash 70/160000, earned 194200, Eco 21; Sol Eco 16; Alien Eco 31. The refineries (large square icons with the harvester glyph) are packed in rows across the plateau, most of them not adjacent to any balterium patch (grey ore symbols). Several sit on the ring of the HQ build radius, and several pairs share one patch.



Screenshot 2, 22:36, whole map. Centauri Eco 25, Sol Eco 17, Alien Eco 33. Centauri (red, top left) and Sol (cyan, bottom right) each sit in one dense blob around the starting HQ; the grey balterium patches across the middle and in the far corners are untouched. Centauri has a second HQ marker centre-left, placed late and away from any patch cluster; Sol has none.



Screenshot 3, Sol base at ground level. Refineries wall to wall, ramps facing all four directions regardless of where the nearest field is; harvesters have to drive around neighbouring refineries to reach the ore, and several of these refineries never receive a harvester trip.

3. Observed: HQ expansion

Over the whole round neither human AI placed an expansion HQ next to an unclaimed patch cluster. The one Centauri expansion in screenshot 2 landed in open ground between the bases at a spot with few patches in reach. The patch groups in the opposite corners (bottom left and top right in screenshot 2) stayed unclaimed for the entire match although both AIs had earned 145k-195k and were banking cash. Expansion, when it happens, is late, rare, and not aimed at resources.

4. Why it happens (from the code review)

Full detail with line references in the linked review. The findings relevant to the two problems:

#	Finding (SilicaCore.dll, namespace Silica.AI; 1.0.9 code, unchanged in 1.0.15)	Effect in play
R1	AIResourcesHandler.GetConstructibleHarvester() returns the cheapest structure that spawns a harvester on completion. For humans that is the Refinery (until an Ultra Heavy Factory exists). Each refinery adds 1 harvester but raises the needed harvester capacity by 2 (num3 = clamp(round(4000/7500), 2, 10) = 2), so the shortfall never closes.	A refinery is requested every think (3 s) while 3000 credits are free. Dominant cause of the spam.
R2	That request targets the ResourceArea nearest world origin (Vector3.zero), not the AI's base, with no served check.	Refineries ring one patch or pile up on the HQ build-radius ring toward the map origin.
R3	The deposit-point path counts patches within 2000 m x MapScale (constant) but can only build within Refinery.MaximumBaseStructureDistance of an HQ. Out-of-reach requests are clamped onto the reach ring.	Refineries on the reach ring that serve nothing; the patch stays unserved and is picked again.
R4	Placement = first valid cell on a square spiral and first valid yaw in the order 0/90/180/270 (useInputRotation: false). The ramp side never enters the decision.	Ramps face a random world axis; the harvester's exit path is ignored.

#	Finding (SilicaCore.dll, namespace Silica.AI; 1.0.9 code, unchanged in 1.0.15)	Effect in play
R5/R6	Served = deposit point within 220 m x MapScale of the moving SignalCenter; split fields are several ResourceAreas.	Large or split fields get a second and third refinery.
H1	An expansion HQ needs tech tier MinimumTeamTier (5 vanilla, 4 on our server) and the expansion code never raises tech priority to get there.	The second HQ comes as late as generic teching allows.
H2	Candidate sites = team start points only.	The spot 'far edge of HQ reach over unclaimed fields' is never evaluated.
H3	Score = resources within 600 m / distance to home; counts patches already served by own refineries or held by the enemy.	Rewards proximity to the main base and already-claimed resources. This is the mis-weighted reward.
H8	A patch out of refinery reach never becomes an expansion trigger for humans (RequestResourceReachTarget only via a relay anchor, which humans do not have).	Out-of-reach fields turn into clamped refineries (R3) instead of an HQ.

Modded values. Refinery and HQ build reach (ConstructionData.MaximumBaseStructureDistance), ResourceCost and MinimumTeamTier are read at runtime, so modded values flow into the placement clamp. WORKED_AREA_REACH (2000 m x MapScale), DEPOSIT_AREA_SERVED_RADIUS (220 m x MapScale), Expansion_SiteResourceRadius (600 m) and ResourceReach_Distance (450 m) are hard-coded and do not scale with a larger reach. With a 1000 m refinery reach the fixed 220 m served radius is smaller than the distance the placement search puts a refinery from a patch, so almost nothing is ever counted as served.

5. Requested behaviour, with our mod as a working reference

The rules below are what a human commander does and what our server-side commander mod Si_RTS_AI (MelonLoader) implements. They are deliberately simple; they hold for vanilla and modded radii because every distance is read from the running game.

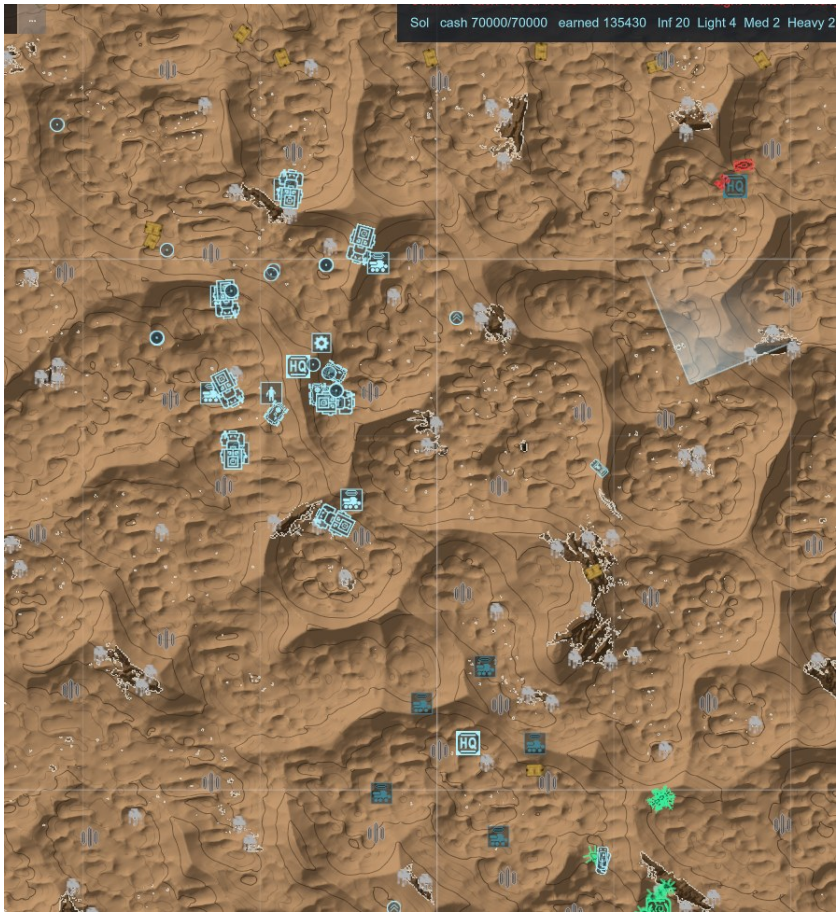
5.1 Refinery: one per patch, as close as possible, ramp toward the field

- Enumerate the team's known non-empty ResourceAreas (ResourceArea.GetKnownResourcesAreas) sorted by distance to the nearest own HQ, so starter patches go up first.
- **Reach.** Only patches within Refinery.MaximumBaseStructureDistance (read from the ConstructionData at runtime) of an own HQ get a refinery. A patch outside that reach is recorded as 'out of reach': it is a demand signal for HQ expansion, never a refinery on the reach ring.
- **Served.** A patch is served if an own refinery, refinery construction site, or refinery landed this round stands within 300 m of it. Areas within 120 m of each other count as one field (split-field guard). Served and pending patches are skipped.
- **Search.** Start the placement search 130 m from the patch centre on the HQ side (between HQ and field is where the ramp belongs), search radius = the refinery's own build reach, and hand the search the wanted facing: yaw = bearing(search point to patch) snapped to 90 degrees, via QueueFirstValidPlacementAroundPoint(..., useInputRotation: true, inputRotation). Every position it returns is then valid for a ramp facing the field.
- **Landing gates** on the returned position: (a) distance to the patch at most 260 m, else reject 'too far'; (b) at least 250 m from any other refinery, else reject 'too close'; (c) the point 30 m in front of the ramp must reach the field on the nav graph, else reject 'ramp cut off' (terrain or a neighbour blocking the ramp side).
- **Construct** with yaw = bearing(landed position to patch) snapped to 90 degrees; fall back to +90 / -90 / 180 only if that rotation fails.
- **Failure memory.** After 3 rejections a patch is marked unserved for the round and becomes an HQ-siting target instead of being retried every think.
- **Budget.** Two refineries before the Research Facility, up to 4 fires per tick afterwards, never more than the team can pay for, hard cap 12.

Measured facts behind this: the ramp faces the structure's yaw direction (Unity yaw 0 = +Z, 90 = +X); human commanders place refineries with yaw within 10-25 degrees of the bearing to the patch; the game accepts refinery rotations only in 90-degree steps; refineries land 170-240 m from the patch centre because of the field's no-build footprint, so any served radius must be sized to that, not to 220 m.

5.2 HQ expansion: score by newly covered, unclaimed patches at the far edge of reach

- Read Refinery and HQ MaximumBaseStructureDistance at scene load; every mask below is built from those values.
- **Candidate cells** (grid over the map): inside HQ reach of an existing own HQ (the new HQ must chain to one) AND outside 2 x refinery reach of every own HQ (the two refinery zones must not overlap, the new HQ has to claim new ground).
- **Value** = (known patches within refinery reach + 400 m of the cell) x (distance to the nearest own HQ) x border factor. The distance factor pushes the pick to the far edge of reach; the border factor = clamp(distance to map edge / refinery reach, 0.25, 1) stops an HQ landing on the map border where half its reach is wasted. Vanilla divides by distance instead, which does the opposite.
- **Demand first.** Before consulting the grid, take the patches the refinery pass gave up on (out of reach or rejected 3 times), nearest first: a field we cannot serve is where the next HQ belongs. This is what makes the second HQ come early and land next to ore.
- **Claims.** Patches already covered by an own refinery are subtracted from the refinery-pressure layer. Patches within an enemy or allied HQ's reach or with an enemy refinery should be excluded from the expansion score too; our mod does not do this yet, it is on our list, and we recommend it for vanilla as well.
- **Execution gates:** one HQ site in flight at a time, 60 s between fires, landing within 300 m of the asked cell and at least 550 m from any other HQ, back off from a cell after 3 fires, and an army reserve must be available to cover the site (the site becomes a standing defence objective until it is functional).
- **Tier gate.** When expansion is wanted and the HQ's MinimumTeamTier blocks it, raise tech priority toward that tier instead of waiting for generic surplus.



Screenshot 4, Sol commanded by our mod (Si_RTS_AI, unfinished) on the same server. Each refinery (square harvester icon) sits on its own balterium patch with the ramp toward it; no refineries on empty ground and no doubles. A second HQ was placed to the south toward the unclaimed patches rather than next to the first one. Sol cash is at the 70000 storage cap in this frame: the mod's spending logic is not finished, but refinery siting and HQ siting are the easy part and already work.

6. What would help us verify a fix

- Per AI think for one human team, log: LastNeededDepositPoints, W, refinery count, harvester capacity vs LastNeededHarvesterCapacity, LastMissingResources, and the ConstructionData plus nearPoint chosen in the harvester-shortfall branch.
- Per placed refinery: distance to the target field, yaw, and the yaw that would have turned the ramp toward the field.
- For expansion: TryPickSite candidates with their scores and the abandon reason.
- We can run any build on our dedicated server (3-way, vanilla or modded radii) and return replays plus the logs above.

Contact: DrMuck (Steam / Silica Discord). Companion review with code references:
https://claude.ai/artifact/NGgmrM5c7UxRWf4Loobwpz